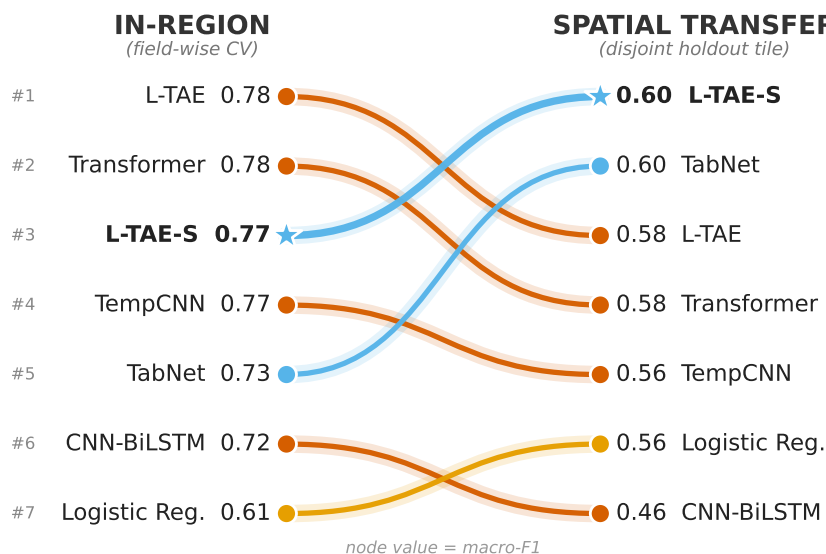


1 Graphical Abstract

2 Sparse Feature Selection, Not Deep Learning Capacity, Drives Out- 3 of-Region Transfer in Crop Classification

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The ranking inverts under spatial transfer

Dense temporal nets lead in-region, then collapse;
sparse feature-selection models rise to the top.

- Dense temporal net ▼ falls
- Linear baseline (robust)
- Sparse feature selection ▲ rises
- L-TAE-S (ours)

6 Sparse Feature Selection, Not Deep Learning Capacity, 7 Drives Out-of-Region Transfer in Crop Classification

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10 Abstract

Crop-type maps from satellite image time series are a backbone of agricultural monitoring, yet the machine-learning models that produce them are almost always trained and tested within a single region, leaving open how they behave when deployed to new areas where no labels were collected. Using a South African Sentinel-2 dataset (optical only), we benchmark twenty-three classical machine-learning and deep-learning models under out-of-region transfer, applying them without any target-region labels or adaptation to an adjacent, non-overlapping holdout tile, and identify the model property that decides success: sparse, axis-aligned feature selection. Models that commit to a few features at a time, gradient-boosted trees, logistic regression, and the attentive TabNet, transfer robustly (macro-F1 0.56–0.60), whereas high-capacity dense temporal networks (CNN-BiLSTM, a Transformer encoder, L-TAE) degrade sharply; a controlled comparison shows that attention capacity is not the lever, sparsity is. We turn this insight into a method: L-TAE-S, a sparse-attention temporal encoder that ties the best model for out-of-region macro-F1 (0.60) at a fraction of the training cost, and whose field-level variant has the smallest train-to-holdout gap of any model. We further show that conventional in-region validation inverts these rankings, the dense networks that lead in-region (macro-F1 0.72–0.78) fall furthest under transfer, which is why models intended for operational crop mapping should be evaluated outside the region they were trained on. Field-level aggregation is a second, substitutable robustness lever, and the most transferable models retain near-full accuracy using only a quarter of the training fields. Together these results give practitioners a concrete design principle for crop classifiers that must be applied to nearby unlabeled farmland.

11 **Keywords:** Crop classification, deep learning, feature selection, remote
12 sensing, Sentinel-2, out-of-region transfer, time-series classification

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1. Introduction

Accurate and timely crop classification is foundational to precision agriculture and to the agencies that monitor food security and agricultural land use, enabling yield forecasting, resource optimization, and food-security assessment (Bégué et al., 2020; Ienco et al., 2017). The Sentinel-2 constellation, with its multispectral richness and roughly five-day revisit, has made dense multi-temporal optical imagery widely available. Yet reliable crop classification remains hard throughout Africa and the Global South: spectral similarity between crops, irregular planting and phenological cycles, persistent cloud cover, and rain-fed dryland management all degrade accuracy (Bégué et al., 2020; Mahlayeye et al., 2024). Classical methods such as Random Forest (RF) and XGBoost have proven effective and efficient on engineered features (Mahlayeye et al., 2024; Mann et al., 2026; Mann and Warner, 2017; Mann et al., 2019), but their performance hinges on hand-crafted feature engineering (Maxwell et al., 2018; Belgiu and Drăguț, 2016). Deep learning, recurrent and convolutional temporal models, reduces reliance on hand-crafted predictors and captures phenological structure that summary statistics miss (Zhu et al., 2017; Rußwurm and Körner, 2018; Pelletier et al., 2019), but its generalizability in complex, low-label settings remains uncertain (Pankajakshan et al., 2025).

A second, and for our purposes critical, gap concerns *how generalization is measured*. Much of the crop-classification literature reports accuracy under random or field-wise cross-validation drawn from a single region (Belgiu and Drăguț, 2016; Garnot and Landrieu, 2020; Pelletier et al., 2019; Rußwurm and Körner, 2018; Ienco et al., 2017). Such protocols hold out individual pixels or fields but not *space*: training and test observations share the same atmosphere, soils, management calendars, and acquisition geometry, so the reported numbers reflect in-distribution fit rather than transfer to new terrain. Because operational maps are produced by applying a model to regions where no labels were collected, this distinction matters, yet studies that re-rank models under a spatially disjoint holdout remain rare (Pankajakshan et al., 2025; Jin et al., 2019; Rustowicz et al., 2019). This differs from the bulk of the transfer-learning literature, which adapts a model using target-domain data: labels, unlabeled imagery, or self-supervised pre-training (Ma et al., 2024; Wang et al., 2023; Mai et al., 2025). Our setting is the no-target-access case (domain generalization): models are trained on source tiles and applied to the holdout directly, with no adaptation, so what determines transfer is a property intrinsic to the model rather than a target-specific tuning step.

The Western Cape of South Africa concentrates the operational challenges, smaller fields, rain-fed dryland systems, heterogeneous planting rotations, and persistent winter cloud cover, while offering an unusually rich, high-quality labeled dataset (Radiant Earth Foundation, 2024). This makes it a realistic test bed: rich enough to train and compare modern architectures fairly, yet hard enough to expose the failure modes that matter in practice. We evaluate deep temporal models (CNN-BiLSTM, TabNet, L-TAE, a Transformer encoder, and TempCNN) and deep patch-based 2D/3D CNNs against classical XGBoost,

LightGBM, and RF. Critically, we evaluate every model twice: under field-wise cross-validation *within* the training region, and on an adjacent, non-overlapping holdout tile that no training pixel touches. We restrict inputs to multi-temporal Sentinel-2 optical imagery alone, to isolate the spectral-time-series signal and maximize applicability where only optical data are available. All code is public (Mann, 2025a).

This study makes five contributions. First, a multi-architecture benchmark (twenty-three classical, deep temporal, and deep patch-based models) showing that in-region versus out-of-region transfer evaluation produce *inverted* rankings on the same data. Second, empirical evidence that models with sparse, axis-aligned feature-selection, the property behind two decades of RF and gradient-boosted-tree dominance, are more stable when moved to a holdout tile, and TabNet inherits it via sparsemax masks. Third, a controlled architectural comparison isolating sparse feature selection rather than attention capacity as the lever, including L-TAE-S, a sparse-attention encoder we introduce that ties TabNet for the best out-of-region macro-F1 and achieves the smallest transfer gap in its field-level variant. Fourth, identification of field-level aggregation as a second, substitutable robustness lever that recovers transfer for dense models but is redundant for sparse ones. Fifth, a data-efficiency analysis under out-of-region transfer showing the most transferable models are also the most data-efficient, holding near-full holdout accuracy on a quarter of the training fields; as a supporting experiment, a controlled comparison of `xr_fresh` time-series features against raw reflectance under out-of-region transfer. This leads to our central question: across classical machine-learning and deep-learning models, which most reliably classifies crops from multi-temporal Sentinel-2 optical imagery when applied, without target-region labels, to an adjacent region outside the training tiles, and what model properties determine whether in-distribution accuracy survives the transfer?

1.1. Related Work

Early crop classification relied on classical learners (RF, SVM, logistic regression, gradient-boosted trees) operating on engineered descriptors (spectral indices, temporal summaries) valued for efficiency, interpretability, and robustness on structured data (Maxwell et al., 2018; Belgiu and Drăguț, 2016; Breiman, 2001; Chen and Guestrin, 2016; Ke et al., 2017), though their performance depends on feature engineering that can limit transferability across agroecological regions (Wardlow et al., 2007; Mahlayeye et al., 2024). Deep models learn representations directly from reflectance sequences: CNNs capture local spectral-spatial pattern (Kussul et al., 2017), and recurrent and attention architectures model seasonal phenology (Ienco et al., 2017; Rußwurm and Körner, 2018; Garnot and Landrieu, 2020; Rußwurm and Körner, 2020). Other designs combine convolutional feature extraction with temporal modeling (Fan et al., 2023; Tufail et al., 2025; Bandar and Coşkunçay, 2024), or use deep backbones as feature extractors feeding classical classifiers (Espejo-Garcia et al., 2020; Koklu et al., 2022; Yang et al., 2020). A recurring assumption is that richer learned representations generalize better than compact engineered ones; whether this

Study area: Western Cape, South Africa

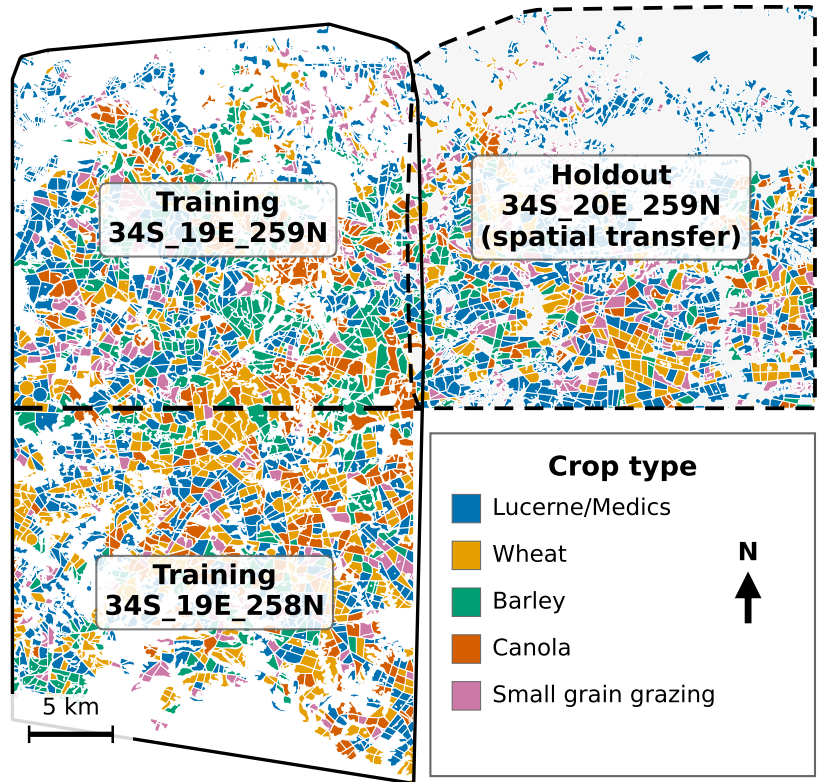


Figure 1: Study area in the Western Cape, South Africa. Two adjacent Sentinel-2 tiles (34S_19E_258N, 34S_19E_259N) form the training region; the immediately adjacent 34S_20E_259N tile (dashed) is held out for out-of-region transfer evaluation. Fields are colored by crop type; the tiles are spatially disjoint, so no pixel or field is shared between training and holdout.

holds under out-of-region transfer, rather than within-region validation, is less established and is a central question here. Architectures that explicitly target cross-scenario generalization (e.g., Cropformer (Wang et al., 2023)) remain the exception and are rarely benchmarked against classical and deep baselines on a common spatially disjoint holdout. We fill this gap with a systematic comparison under *both* protocols, and additionally evaluate `xr_fresh` as an automated time-series feature-extraction framework in a crop-classification setting (Mann, 2025b).

111 2. Data

112 The study uses a labeled field dataset distributed through Radiant Earth’s
113 MLHub (Radiant Earth Foundation, 2024) and used as the basis of the AI4FoodSecurity
114 (South Africa) challenge (AI4EO, 2021), in an agriculturally productive part
115 of the Western Cape (Fig. 1). Two adjacent Sentinel-2 tiles (34S_19E_258N,
116 34S_19E_259N) form the training region (4,150 labeled fields over 622 km²; mean
117 field 15.0 ha); a third, immediately adjacent tile (34S_20E_259N) is reserved as
118 a holdout for out-of-region transfer evaluation (2,417 fields over 282 km²). The
119 tiles do not overlap, so no pixel or field is shared, and transfer must contend
120 with different soils, geometries, and acquisition conditions. Because the holdout
121 lies within the same agroecological zone and growing calendar, this is a rela-
122 tively *local* shift, and the gaps we report are best read as a conservative lower
123 bound on degradation under broader transfer. Throughout, “in-region” denotes
124 a field-wise train/validation/test split within the training region; “out-of-region”
125 (OOR) or “holdout” denotes predictions on the adjacent, non-overlapping tile,
126 made without any target-region labels or adaptation. We use “out-of-region”
127 rather than the broader “out-of-region transfer” because the holdout tests a
128 local shift, not transfer across agroecological zones or years.

129 We classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley,
130 and small-grain grazing. The classes are heavily imbalanced; lucerne/medics
131 is 43% of training fields and 56% of the holdout (Supplementary Material,
132 Fig. S1(a)), which motivates macro-F1 as the primary metric (Sec. 3.3) and
133 class-imbalance handling during training. Spectrally, wheat and barley overlap
134 strongly, the dominant source of persistent confusion (Supplementary Mate-
135 rial, Fig. S2(b)). Sentinel-2 Level-2A surface reflectance was acquired for the
136 January–December 2017 growing season, cloud-masked with `s2cloudless` plus
137 a shadow mask, and composited monthly. Six features per month are used:
138 bands B2, B6, B11, B12, plus EVI and Hue, with EVI and Hue computed per
139 scene before compositing, chosen to span the most informative spectral regions
140 while limiting redundancy among correlated bands (Mann et al., 2026; Tufail
141 et al., 2025). June 2017 yielded no usable imagery and May was discarded
142 for cloud, leaving 10 monthly observations, so each pixel is a 60-dimensional
143 time series (6 features $\times$ 10 months). Field polygons were digitized from sub-
144 meter aerial photography and carry a single seasonal crop label (Radiant Earth
145 Foundation, 2024); all predictions are made at the field level.

146 3. Methodology

147 We compare two model families: classical machine learning on engineered
148 `xr_fresh` features (9 models) and deep learning (14), the latter comprising deep
149 temporal models on per-pixel or per-field time series (12) and deep patch-based
150 models on image patches (2). To prevent leakage, all train/validation/test split-
151 ting is performed at the field level with a fixed random seed, so no field’s pixels
152 straddle partitions; pixel- and patch-level predictions are aggregated to a field

label by majority vote (or, for probabilistic models, mean pooling). Computational cost across paradigms is reported in the Supplementary Material (Sec. S-C).

3.1. Classical Machine Learning

We automate feature extraction with the `xr.fresh` toolkit, computing a set of statistical and temporal descriptors (e.g., autocorrelation, linear trend, day-of-year of the seasonal maximum, quantiles, complexity) from each pixel’s 10-month profile, yielding a 174-dimensional per-pixel vector (Mann et al., 2026; Mann, 2025b); the full feature list is in the Supplementary Material (Sec. S-A, Table S1). Four baselines: logistic regression, RF (300 trees), LightGBM, and XGBoost (James et al., 2013; Breiman, 2001; Ke et al., 2017; Chen and Guestrin, 2016; Friedman, 2001) are trained on this representation, all of them tree- or margin-based with built-in feature selection or shrinkage; standard formulations are reproduced in the Supplementary Material (Sec. S-A). We also report soft-voting and stacked ensembles (RF/XGBoost/LightGBM, with logistic-regression meta-learner), an Optuna-tuned field-level XGBoost, and a SMOTETomek-resampled variant; the *Stacking* and *SMOTE Stacked* pair differs only in resampling, isolating the effect of class rebalancing. At the field level, pixel features are mean-pooled within each polygon before classification (Peña-Barragán et al., 2011).

3.2. Deep Learning Architectures

At the pixel level each observation is a 60-feature temporal sequence. Although the pixel models train on millions of pixels, pixels within a field are highly autocorrelated and share one label, so the number of statistically independent units is closer to the number of fields; field-wise splitting keeps evaluation honest at the field scale. We evaluate, as five-seed ensembles, a CNN-BiLSTM trained with weighted focal loss; TabNet (Arik and Pfister, 2021; Muruganathan et al., 2024), which treats the observations as tabular input and, at each decision step, uses an attentive transformer to produce a *sparse* feature mask via sparsemax, selecting a handful of features and ignoring the rest; the L-TAE (Garnot and Landrieu, 2020), which applies a single learned query over the temporal sequence; a standard multi-head Transformer encoder (Vaswani et al., 2017; Rußwurm and Körner, 2020); and TempCNN (Pelletier et al., 2019). Detailed equations for these standard models, and the field-level voting rule, are in the Supplementary Material (Sec. S-A). The Transformer is included as an informative dense-attention comparison: it increases attention capacity relative to L-TAE while retaining no explicit sparse feature-selection mechanism. Deep patch-based models (a multi-channel 2D CNN and a 3D CNN) tile each field into 100 m patches resampled to 128×128 and are likewise trained as five-seed ensembles; details in the Supplementary Material (Sec. S-A).

3.2.1. Sparse-Attention L-TAE (L-TAE-S)

To test whether a tree-like, sparse inductive bias can be grafted onto a temporal-attention encoder, we introduce L-TAE-S, which inserts a TabNet-inspired channel-selection gate before the attention heads (Fig. 2). For each of the $n_{\text{head}} = 16$ heads, a per-head linear projection of the embedded, positionally-encoded sequence ($d_{\text{model}} = 128$, key dimension $d_k = 8$) is passed through *sparsemax* (Martins and Astudillo, 2016) to produce a sparse, instance-dependent mask over embedding channels. Formally, for head h and time step $\mathbf{z}_t \in \mathbb{R}^E$,

$$\mathbf{m}_{h,t} = \text{sparsemax}(\mathbf{P}_{h,t} \odot \text{BN}(W_h \mathbf{z}_t)), \quad \tilde{\mathbf{z}}_{h,t} = \mathbf{m}_{h,t} \odot \mathbf{z}_t,$$

$$\mathbf{P}_{h+1,t} = \mathbf{P}_{h,t} \odot \max(\gamma - \mathbf{m}_{h,t}, 0), \quad \mathbf{P}_{1,t} = \mathbf{1};$$

head h then forms its keys and values from $\tilde{\mathbf{z}}_{h,t}$ before applying L-TAE attention. Because *sparsemax* projects onto the probability simplex, most channels of $\mathbf{m}_{h,t}$ are exactly zero, and the relaxation prior $\mathbf{P}_{h,t}$ (coefficient $\gamma = 1.5$) discourages successive heads from re-selecting the same channels. These two mechanisms are the sole source of sparsity. No explicit sparsity penalty is used, because an ℓ_1 penalty on a sparsemax mask is constant ($\|\mathbf{m}_{h,t}\|_1 = 1$) and has zero gradient. This single block, the only architectural difference from the L-TAE, grafts a tree-like sparse feature-selection bias onto the attention backbone while keeping its temporal modeling, mirroring TabNet’s sparsemax masks (Arik and Pfister, 2021). We evaluate L-TAE-S at both the pixel and field level as a five-seed ensemble under the same weighted-focal objective and field-aggregation protocol as the other temporal networks.

3.3. Performance Metrics

Because the classes are severely imbalanced, plain accuracy is misleading. We adopt macro-F1 ($F1_m$, the unweighted mean of per-class F1) as the primary metric, since it weights every crop equally and penalizes failure on the minority cereals where models differ most. We report three further metrics: Cohen’s Kappa; weighted F1 (wF1, the support-weighted mean of per-class F1); and cross-entropy (Xent), the field-level scoring metric of the AI4FoodSecurity challenge (AI4EO, 2021) from which the dataset is drawn (lower is better, computed on hard label predictions). Formal definitions are in the Supplementary Material (Sec. S-B).

3.4. Quantifying Uncertainty

We quantify two sources of variability. For sampling uncertainty in the finite holdout, we apply a nonparametric bootstrap over the 2,417 holdout fields: for each model we resample its per-field predictions with replacement $B = 10,000$ times and recompute macro-F1, reporting the 2.5th–97.5th percentile interval as a 95% confidence interval (CI); For between-model gaps, we compute paired bootstrap differences by resampling holdout fields and subtracting model scores within each bootstrap replicate. For optimization noise, the per-seed standard deviation of OOR macro-F1 across the five ensemble members is 0.015–0.037

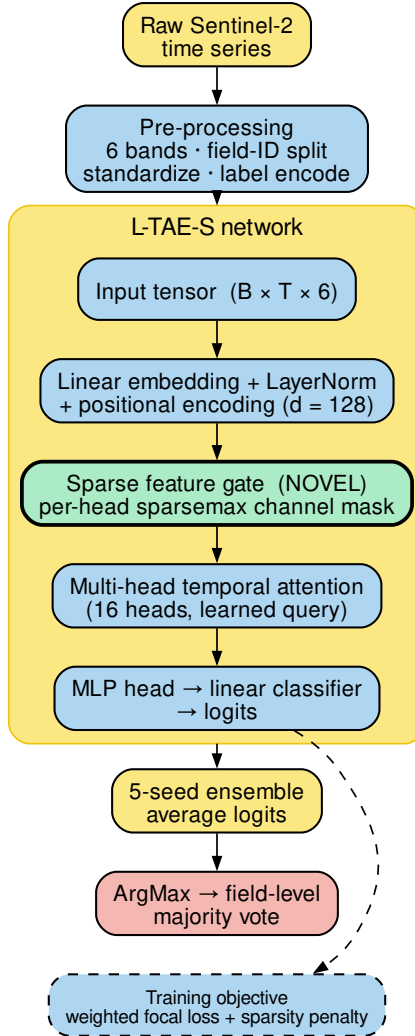


Figure 2: Architecture of the sparse-attention L-TAE-S. Before the multi-head temporal attention, a TabNet-style sparse feature gate (highlighted) applies a per-head **sparsemax** mask that selects a small subset of embedding channels, with a relaxation prior ($\gamma = 1.5$) discouraging heads from re-selecting the same channels. This block is the only architectural difference from the L-TAE; five seeds are ensembled and aggregated to field level.

233 for the headline temporal models, which is why we report five-seed ensembles
234 throughout. Both sources are small relative to the ranking effects below: every

field-bootstrap CI spans only $\approx \pm 0.02$ – 0.03 macro-F1, so we treat OOR macro-F1 differences below ≈ 0.05 as not statistically distinguishable.

4. Results

We evaluate every model under two protocols: 1) in-region field-wise validation and 2) out-of-region transfer (OOR) to the adjacent, non-overlapping holdout tile. All reported metrics are field-level. Pixel- and patch-level models are aggregated to field labels, whereas field-level models are trained and evaluated on field-aggregated inputs

4.1. In-Region Validation Overstates Performance and Misranks Models

Table 1 reports both protocols for all models, ranked by OOR macro-F1, with the gap $\Delta = F1_m^{\text{OOR}} - F1_m^{\text{in-region}}$. Two facts stand out. First, *every* model loses accuracy under transfer, but unevenly: from -0.04 (field-level L-TAE-S) to -0.26 (CNN-BiLSTM at the pixel level). Second, this unevenness reorders the models. In-region the dense temporal pixel nets cluster at the top (L-TAE 0.78, Transformer 0.78, L-TAE-S 0.77, TempCNN 0.77, CNN-BiLSTM 0.72); a practitioner choosing by in-region F1 would pick from this cluster. On the holdout the cluster fractures: the two in-region leaders (L-TAE pixel and the Transformer, $\Delta = -0.20$) drop to 0.58, TempCNN to 0.56, and CNN-BiLSTM ($\Delta = -0.26$) to 0.46, among the lowest absolute scores of any model.

The winners under transfer are a different group. The best single model is the TabNet pixel ensemble ($F1_m = 0.60$, $\kappa = 0.54$), which also posts the lowest holdout cross-entropy (4.42) of any model, tied on macro-F1 by both sparse-attention L-TAE-S variants (0.60). The most *robust* models, those with the smallest gaps, are the sparse learners: field-level L-TAE-S posts the single smallest gap ($\Delta = -0.04$), just ahead of logistic regression (-0.05) and the pixel-level tree learners. That L-TAE-S, a deep temporal attention network whose only twist is a learned gate restricting each head to a small channel subset, sits atop the robustness ranking is itself evidence for our thesis. The field bootstrap confirms which differences are real: the best transferers beat CNN-BiLSTM by a wide, significant margin (TabNet over CNN-BiLSTM pixel, $\Delta = 0.14$, 95% CI $[0.11, 0.18]$, $p < 0.001$), so the inversion is not holdout noise; at the very top the leaders (TabNet and both L-TAE-S variants) are statistically indistinguishable (all 0.60; TabNet vs. L-TAE-S, $p = 0.80$). We therefore do not claim a single best model; we claim that a sparse-feature-selection group occupies the top of the OOR ranking while the dense temporal nets prized by in-region validation fall significantly below it.

Table 1: In-region versus out-of-region transfer performance of all twenty-three models.

Model	Level	Features	In-reg		Out-of-region (OOR)				
			$F1_m$	Δ	$F1_m$	95% CI	κ	wF1	Xent
TabNet (raw)	pixel	raw	0.73	-0.13	0.60	.58-.63	0.54	0.70	4.42
L-TAE-S	field	raw	0.64	- 0.04	0.60	.58-.62	0.52	0.69	4.77
L-TAE-S	pixel	raw	0.77	-0.17	0.60	.57-.62	0.53	0.70	4.63
L-TAE	field	raw	0.67	-0.07	0.59	.57-.62	0.47	0.67	5.51
L-TAE	pixel	raw	0.78	-0.20	0.58	.55-.60	0.49	0.68	5.20
Transformer	pixel	raw	0.78	-0.20	0.58	.55-.60	0.50	0.69	4.94
XGBoost	field	xr_fresh	0.70	-0.13	0.56	.54-.59	0.47	0.68	4.89
TempCNN	pixel	raw	0.77	-0.21	0.56	.54-.59	0.49	0.68	5.09
LightGBM	pixel	xr_fresh	0.62	-0.06	0.56	.54-.59	0.47	0.66	4.85
Logistic Regression	pixel	xr_fresh	0.61	-0.05	0.56	.54-.59	0.47	0.67	4.93
LightGBM	field	xr_fresh	0.68	-0.12	0.56	.53-.58	0.45	0.66	4.99
Voting	pixel	xr_fresh	0.69	-0.13	0.56	.53-.58	0.48	0.66	4.85
Random Forest	pixel	xr_fresh	0.65	-0.10	0.55	.52-.57	0.45	0.65	5.07
XGBoost	pixel	xr_fresh	0.64	-0.09	0.54	.52-.57	0.47	0.66	5.01
TempCNN	field	raw	0.67	-0.14	0.54	.51-.56	0.43	0.64	6.07
Stacking	field	xr_fresh	0.64	-0.11	0.53	.51-.56	0.48	0.67	5.32
CNN-BiLSTM	field	raw	0.64	-0.14	0.51	.48-.53	0.33	0.54	7.86
3D CNN	patch	raw	0.70	-0.20	0.50	.48-.53	0.29	0.54	7.98
SMOTE Stacked	field	xr_fresh	0.69	-0.21	0.49	.46-.51	0.43	0.63	5.37
CNN-BiLSTM	pixel	raw	0.72	-0.26	0.46	.44-.48	0.32	0.56	7.73
Multi-Channel 2D CNN	patch	raw	0.63	-0.18	0.45	.42-.47	0.30	0.54	7.62
TabNet (raw, field-avg)	field	raw	0.62	-0.19	0.43	.41-.46	0.31	0.56	7.35
TabNet (xr_fresh)	field	xr_fresh	0.51	-0.15	0.37	.34-.40	0.25	0.53	6.13

Ranked by OOR macro-F1. In-region uses field-wise validation; OOR uses the disjoint holdout tile (34S.20E.259N). Δ is the gap; the 95% CI is the percentile bootstrap on OOR macro-F1 from 10,000 resamples of the 2,417 holdout fields. Xent is the AI4FoodSecurity field-level scoring metric (lower is better). *Features* marks the automated **xr_fresh** statistics (174 features) versus raw reflectance. Bold marks the best value per column.

4.2. Pixel Versus Field Level: Aggregation as Variance Reduction

Several architectures were run at both the pixel and the field level (averaging per-pixel sequences within a field before classification). Table 2 shows that field-level aggregation acts as a variance-reduction regularizer for the *dense* temporal networks: L-TAE’s gap shrinks from -0.20 to -0.07 , and CNN-BiLSTM, the worst pixel-level transferer, recovers most, its gap nearly halving from -0.26 to -0.14 ($F1_m$ $0.46 \rightarrow 0.51$). The sparse-attention L-TAE-S follows the same dense-backbone pattern despite its channel gating: its gap shrinks from -0.17 to -0.04 while OOR $F1_m$ holds at 0.60, so the response to aggregation appears to depend on both the temporal backbone and the input granularity; the sparse gate does not eliminate the benefit of field-level aggregation. TabNet is the informative exception: aggregating its raw inputs to the field level *hurts* ($F1_m$ $0.60 \rightarrow 0.43$), because its robustness already derives from sparse feature selection and field-level training starves it of the per-pixel observation volume it exploits. The tree learners transfer well at either level, and their pixel $\rightarrow$ field change is essentially neutral under transfer (XGBoost $0.54 \rightarrow 0.56$). In short, dense models need aggregation to generalize; sparse, tree-like models do not.

Table 2: Pixel- versus field-level out-of-region transfer.

Architecture	Pixel		Field		Field input
	F1 _m	Δ	F1 _m	Δ	
L-TAE-S	0.60	−0.17	0.60	−0.04	raw-avg
L-TAE	0.58	−0.20	0.59	−0.07	raw-avg
XGBoost	0.54	−0.09	0.56	−0.13	xr_fresh
LightGBM	0.56	−0.06	0.56	−0.12	xr_fresh
TempCNN	0.56	−0.21	0.54	−0.14	raw-avg
CNN-BiLSTM	0.46	−0.26	0.51	−0.14	raw-avg
TabNet	0.60	−0.13	0.43	−0.19	raw-avg

Architectures run at both granularities, ranked by field-level OOR F1_m. The *Field input* column gives the field-level input (**xr_fresh** statistics for the trees, field-averaged raw sequence for the deep models). Field aggregation helps the dense temporal nets, is roughly neutral for the tree learners, and degrades TabNet. Bold marks the best value per column.

4.3. Per-Class Performance and Persistent Confusions

Figure 3 breaks accuracy down crop by crop on the holdout for four representative models: the best transferer (TabNet), our sparse-attention L-TAE-S, its dense counterpart L-TAE, and CNN-BiLSTM (the largest drop under transfer). The errors are highly uneven. All three robust models classify the dominant lucerne/medics and the distinctive canola well (TabNet F1 0.88 and 0.81; L-TAE-S 0.87 and 0.76; L-TAE 0.84 and 0.66), but every model struggles with the cereals and small-grain grazing: even TabNet reaches only 0.55 on barley, 0.48 on wheat, and 0.30 on small-grain grazing. Adding the sparsity gate (L-TAE → L-TAE-S) on the same backbone improves the easier classes (canola 0.66 → 0.76) while leaving the hard cereals essentially unchanged (barley 0.50 → 0.50, wheat 0.48 → 0.47). CNN-BiLSTM, by contrast, loses accuracy on the easy classes too (lucerne/medics 0.71, canola 0.65), showing its failure is a general inability to transfer, not a minority-class problem alone. The wheat–barley confusion is consistent with their spectral overlap and appears across every model.

4.4. Data Efficiency Under Out-of-Region Transfer

The most transferable models are also the most data-efficient. We retrain a representative set on random 75%, 50%, and 25% subsets of the training fields and re-evaluate on the holdout; because a single subsample can be unrepresentative, we repeat each fraction over five independent draws and plot the mean (Fig. 4). Averaged this way, the deep temporal models are essentially flat from full data down to a quarter of the fields (L-TAE and L-TAE-S vary by under 0.02 macro-F1 across fractions), while the tree, linear, and TabNet baselines lose only ~0.03–0.05—modest given a four-fold cut in training data. The per-fraction wiggles seen in any single split are within draw-to-draw sampling variance: L-TAE-S, whose sparse channel gate makes it the most sensitive to *which* fields are sampled, swings ±0.06 macro-F1 at the 50% fraction (versus ≤ 0.02 for the robust learners), so the apparent 50% “dip” of any one draw is sampling noise rather than a data-efficiency effect. Holdout accuracy is therefore

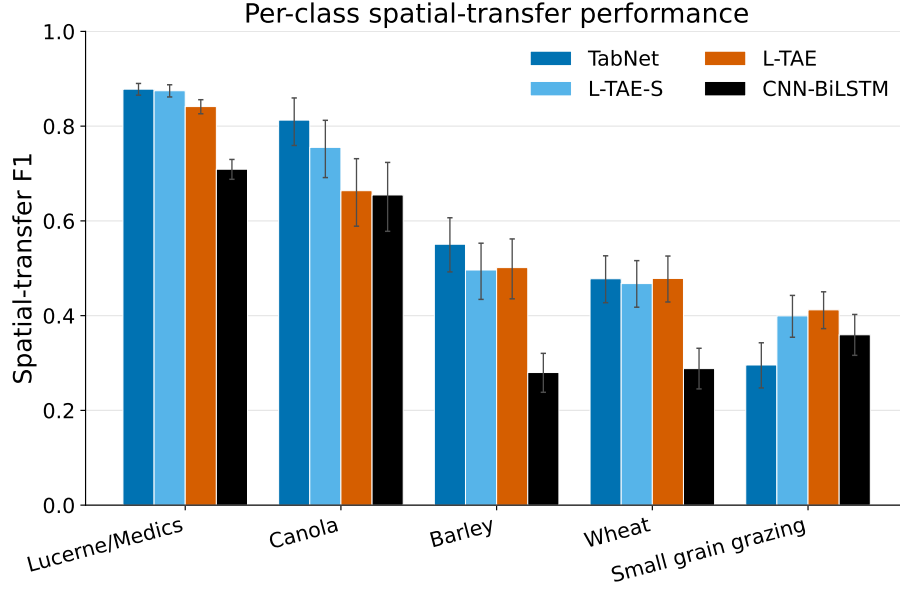


Figure 3: Per-class F1 on the holdout region for four representative models (TabNet, L-TAE-S, L-TAE, and CNN-BiLSTM). TabNet, L-TAE-S, and L-TAE classify lucerne/medics and canola well; the cereals (barley, wheat) and small-grain grazing concentrate the errors. CNN-BiLSTM’s accuracy drops on every class. Error bars are 95% bootstrap CIs over holdout fields.

322 limited far more by out-of-region transfer than by training-set size, reinforcing
 323 that what governs holdout accuracy is the model’s inductive bias rather than
 324 the volume of training data.

325 4.5. Raw Reflectance versus *xr_fresh*

326 The best-transferring classical learners in Table 1 are exactly those fed
 327 *xr_fresh* statistics, so representation and model family are confounded there.
 328 To separate them, we trained a single XGBoost learner once on raw monthly
 329 reflectance and once on the per-pixel *xr_fresh* statistics, holding the FID-wise
 330 splits, preprocessing, and field-level mean pooling fixed (Table 3). The two rep-
 331 resentations are essentially indistinguishable under transfer ($F1_m = 0.59$ raw
 332 vs. 0.56 *xr_fresh*, a difference within the holdout bootstrap interval of Table 1;
 333 equal weighted-F1 and κ), and raw reflectance is, if anything, marginally ahead
 334 in-region as well. The *xr_fresh* representation therefore yields no measurable
 335 transfer-accuracy gain over raw reflectance for the trees; its value is operational
 336 - a compact, fast-to-fit, agronomically interpretable vector that trains in sec-
 337 onds (Supplementary Material, Sec. S-C), rather than an accuracy edge. This
 338 is, to our knowledge, the first controlled comparison of *xr_fresh* against raw
 339 reflectance under out-of-region transfer.

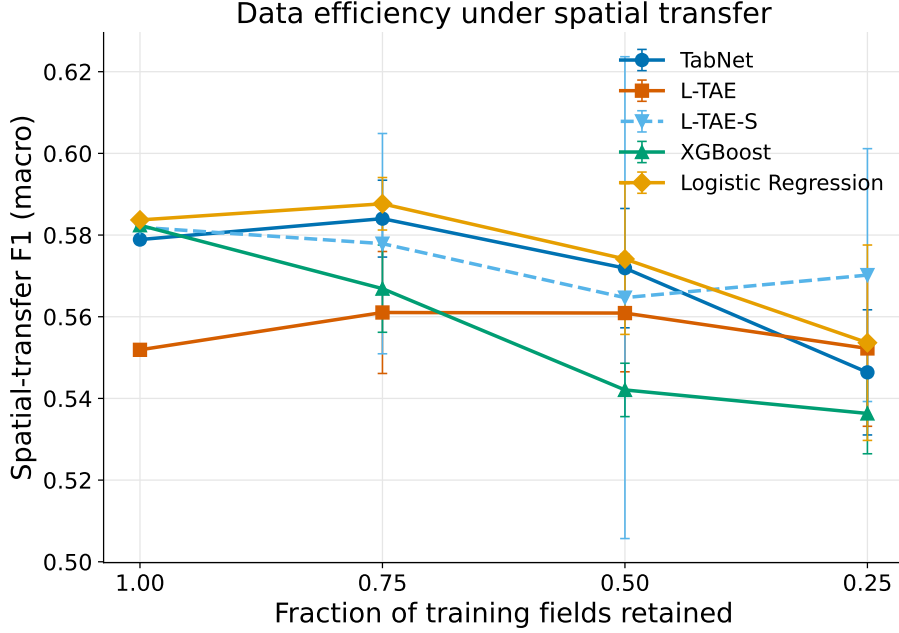


Figure 4: Data efficiency under out-of-region transfer: macro-F1 versus fraction of training fields retained, single training seed, each point the mean over five independent subsample draws (error bars ± 1 s.d.). The deep temporal models (L-TAE, L-TAE-S) are nearly flat to a quarter of the fields; the tree, linear, and TabNet baselines decline modestly. Per-fraction variation is within draw-to-draw sampling noise, largest for L-TAE-S (± 0.06 at 50%), so single-split “dips” are not real. Absolute values run ~ 0.02 – 0.05 below the five-seed levels of Table 1 because a single training seed is used.

Table 3: Raw reflectance versus `xr_fresh` for a single fixed XGBoost learner.

Representation	Split	$F1_m$	wF1	κ
Raw reflectance	In-region	0.73	0.80	0.71
<code>xr_fresh</code>	In-region	0.72	0.76	0.68
Raw reflectance	Out-of-region	0.59	0.69	0.50
<code>xr_fresh</code>	Out-of-region	0.56	0.69	0.50

Identical FID-wise splits, preprocessing, and field-level mean pooling; only the input representation differs. Metrics are field-level macro-F1 ($F1_m$), weighted F1 (wF1), and Cohen’s κ .

5. Discussion

The conclusion about which model is “best” depends almost entirely on whether evaluation is in-region or under out-of-region transfer, and a model’s transfer behavior is predictable from its inductive bias.

5.1. Why Tree-Like Inductive Bias Transfers

The clearest pattern is that transfer is strongly associated with inductive bias, and not simply with model family or attention capacity. Figure 5 groups every model by inductive bias and plots its OOR gap: linear models lose the least (mean $\Delta = -0.05$), tree-based models little (mean -0.11), the sparse-attention L-TAE-S sits essentially with the trees (mean -0.11 across its variants), while dense temporal/patch nets lose the most (mean -0.21) and SMOTE-augmented stacking is similarly fragile (-0.21). RF and gradient-boosted trees have been the remote-sensing workhorse for two decades not because they top in-region accuracy (dense nets often do) but because they survive when a model is applied to an unseen region (the operational case) (Belgiu and Drăguț, 2016; Maxwell et al., 2018). A tree decides one feature at a time, choosing the few features that matter for each split and ignoring the rest; this sparsity may reduce reliance on region-specific quirks of phenology, soil, and atmosphere that pull richer models off course. Independent corroboration spans settings: a 3D-CNN collapses under cross-sensor hyperspectral transfer while a sparse SVM holds (Pankajakshan et al., 2025); RF matches or exceeds a dense CNN-LSTM on label-scarce Ghana fields (Rustowicz et al., 2019); a maize model drops 16 points moving across East Africa (Jin et al., 2019); and bagged/boosted trees out-transfer an elaborate instance-reweighting scheme from the U.S. to northwest China (Mai et al., 2025). TabNet, the best transferer here, is a neural network built to behave like a tree: a small learned gate picks a handful of features at each step (Arik and Pfister, 2021), so it clusters with the trees and inherits their transferability. By contrast, CNN-BiLSTM, TempCNN, L-TAE, and the Transformer consume the whole temporal sequence at once with no feature selection. At the pixel level this leaves them fitting the training region tightly: they post the largest in-region-to-holdout gaps in the benchmark, with CNN-BiLSTM dropping furthest. This is consistent with evidence that the dates such models attend to are themselves contingent on the training region (Obadic et al., 2022). Dense attention is not intrinsically poor under out-of-region transfer: the pixel-level L-TAE loses 0.20 macro-F1, but the field-level L-TAE reduces this gap to 0.07 and the other temporal nets (Sec. 4.2), so it is the pixel-level dense models specifically that transfer worst.

The Transformer is a particularly clean control: although it deploys the most expressive attention of any model we test, far richer than L-TAE’s single query, at the pixel level it transfers no better than the pixel-level L-TAE (both $F1_m = 0.58$, $\Delta = -0.20$). Within the architectures evaluated, increasing attention capacity alone did not improve transfer, whereas adding feature *selection* does: L-TAE-S and TabNet, the two models that explicitly select features before computing with them, are the best transferers. This isolates feature selection—not the attention mechanism, the temporal backbone, or model size—as the property governing transfer. The L-TAE-S experiment adds that this is *not* a tree-versus-deep tradeoff: the same selection step that drives tree splits can be grafted onto a dense architecture at the cost of a single block, and the payoff is large; L-TAE-S ties TabNet for the best holdout accuracy. That parity is not

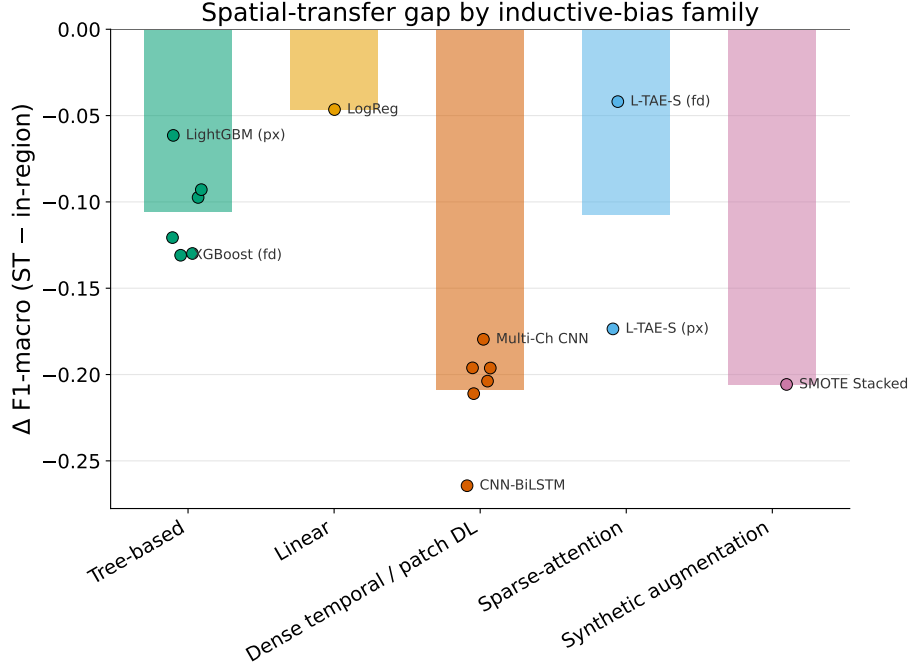


Figure 5: Out-of-region transfer gap (Δ macro-F1) grouped by inductive-bias family. Sparse/tree-like and linear models transfer (small $|\Delta|$); dense temporal/patch nets and synthetic augmentation overfit the training region. L-TAE-S, which adds sparse feature selection to a dense attention backbone, tracks the tree-based models. Bars are family means; points are individual models.

accuracy alone: at the pixel level L-TAE-S reaches TabNet’s best-in-class hold-out accuracy in roughly 40% of the per-seed training time (24 vs. 61 min), and its field-level variant is both faster and more accurate than any field-aggregated TabNet (Supplementary Material, Sec. S-C). The next generation of operational crop classifiers need not choose between the expressivity of deep learning and the transfer-robustness of trees.

5.2. Two Robustness Levers

Two caveats sharpen rather than weaken the story, and together they point to how we can choose the best models. First, it is *pixel-level* TabNet that wins; the field-aggregated TabNet variants are among the worst transferers (Table 1), because aggregation removes the per-pixel observation volume the sparse model exploits. Second, dense temporal attention transfers well *when* applied at the field level: L-TAE’s pixel-level gap shrinks substantially under aggregation, and L-TAE-S ends with an even smaller gap (Table 2). These point to two distinct, partly substitutable directions. The first lever is built-in feature selection. The model itself picking a few features at a time (trees, TabNet, and now L-TAE-S);

405 this is intrinsic robustness paid for at design time. The second is variance re-
 406 duction imposed from outside via field-level aggregation, which averages away
 407 region-specific pixel noise before the classifier sees it. Which provides external
 408 robustness available even when the architecture is fixed. A model with the first
 409 lever gains nothing, and may lose observation volume, from the second; this
 410 is exactly why TabNet prefers pixels. These two properties can be combined.
 411 L-TAE-S adds a sparse gate to a temporal-attention backbone and then benefits
 412 from field aggregation, ending with the smallest transfer gap in the benchmark.
 413 Practitioners can read this as a flowchart: if free to choose or modify the model,
 414 build feature selection in (lever 1); if constrained to an off-the-shelf dense archi-
 415 tecture, aggregate predictions to the field level (lever 2). Two further controlled
 416 comparisons confirm the ranking is not an artifact of ensembling or imbalance
 417 handling; both shift absolute scores but never reorder the robust and fragile
 418 groups, with the full tables in the Supplementary Material (Sec. S-D). Seed-
 419 ensembling gives only a small, monotone lift that leaves the ordering intact. A
 420 single-seed L-TAE-S ($F1_m = 0.56$) already exceeds the five-seed CNN-BiLSTM
 421 and TabNet field ensembles (Table S4). And the imbalance-handling objec-
 422 tive moves absolute scores but not the ranking: the robust L-TAE-S is almost
 423 invariant across five loss and sampler settings ($F1_m = 0.59$ – 0.60), whereas the
 424 fragile CNN-BiLSTM swings widely (0.41 – 0.56) and even at its best stays below
 425 L-TAE-S’s worst (Table S5).

426 5.3. Automated Features, Patch Models, and Persistent Difficulty

427 The classical pipelines paired with `xr_fresh` were not only robust but cheap,
 428 transferring as well as the best deep models (OOR $F1_m$ 0.55 – 0.56) while training
 429 in minutes rather than hours (Supplementary Material, Sec. S-C); the controlled
 430 comparison in Sec. 4.5 confirms that this value is operational rather than an ac-
 431 curacy edge over raw reflectance. Patch-based convolutional models did not
 432 exceed pixel-level models and transferred poorly (3D CNN $F1_m = 0.50$; multi-
 433 channel 2D CNN 0.45): at 10 m resolution the native 100 m patches are only
 434 $\sim 10 \times 10$ pixels, so the spatial context is limited and resampling adds inter-
 435 polation rather than detail, and a dense convolutional representation offers no
 436 transfer-protective sparsity. Finally, small-grain grazing, barley, and wheat are
 437 the hardest classes for every model (Fig. 3) and account for most accuracy lost
 438 under transfer; that even robust models struggle here, and that CNN-BiLSTM
 439 loses accuracy on every class, indicates the limitation is partly intrinsic spectral
 440 similarity and partly the transfer problem itself, not class imbalance alone.

441 5.4. Limitations

442 The study uses a single growing season (2017) and a single holdout tile that
 443 is spatially *adjacent* to the training tiles within the same agroecological zone.
 444 This is a relatively local shift, so the gaps we report most likely *understate*
 445 the degradation that fully cross-regional or cross-year transfer would produce;
 446 our numbers are best read as a conservative lower bound, and multi-region,
 447 multi-year holdouts are needed to confirm the inductive-bias ordering is stable

under larger shifts. Our conclusions should therefore be read as applying to out-of-region transfer between neighboring tiles, not to cross-ecozone or cross-year transfer. The labels derive from a public dataset rather than our own campaign, and we use Sentinel-2 optical data only; a deliberate scoping choice that nonetheless limits the achievable ceiling for optically similar cereals, making SAR fusion a clear next step.

6. Conclusion

In-region validation, the protocol the crop-classification literature reports almost universally, misranks twenty-three Sentinel-2 classifiers when they are applied to an adjacent, non-overlapping holdout tile. Dense temporal networks that lead under conventional evaluation fall sharply under transfer, the worst of them (CNN-BiLSTM) landing near the bottom, while parsimonious models such as tree ensembles, logistic regression, and a TabNet ensemble retain rank and accuracy. The mechanism is the same property that made tree ensembles the remote-sensing default for two decades: a sparse, axis-aligned feature-selection bias that commits to a few robust signals and ignores the rest. TabNet inherits it through sparsemax masks; a Transformer with far greater attention capacity does not, and transfers no better than the lightweight L-TAE, pointing at sparsity, not capacity, as the lever. Grafting that lever onto an attention model (L-TAE-S) ties TabNet for the best transfer at a fraction of its training cost and, at the field level, posts the smallest in-region-to-holdout gap of any model. Field-level aggregation is a second, partly substitutable lever that helps with sparsity in L-TAE-S but hurts TabNet. We therefore recommend the following: evaluate on an out-of-region holdout before declaring one architecture better than another; prefer models with a built-in feature-selection bias when mapping nearby unlabeled areas; and, if dense temporal models are required, aggregate their predictions to the field level. The conditions that make the Western Cape difficult are the conditions under which agricultural remote sensing must operate across much of the Global South, and evaluating models the way they are actually deployed, applied to regions where no labels were collected, is what will make remote sensing deliver where it matters most.

CRedit authorship contribution statement

Michael L. Mann: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Sairam Venkatachalam:** Software, Formal analysis, Writing – original draft, Investigation. **Disha Kacha:** Software, Formal analysis, Investigation, Writing – original draft. **Devarsh Sheth:** Software, Formal analysis, Investigation, Writing – original draft. **Ryan Engstrom:** Supervision, Methodology, Writing – review & editing. **Amir Jafari:** Methodology, Methodology, Supervision, Writing – review & editing.

488 **Ethics statement**

489 This article does not contain any studies with human participants or animals
490 performed by any of the authors. The study is based solely on publicly avail-
491 able satellite imagery (Sentinel-2) and field-boundary crop labels, so no ethical
492 approval was required.

493 **Declaration of competing interest**

494 The authors declare that they have no known competing financial interests or
495 personal relationships that could have appeared to influence the work reported
496 in this paper.

497 **Data availability**

498 The processed dataset supporting this study is openly available on IEEE
499 DataPort at <https://dx.doi.org/10.21227/n69k-r458> (DOI: 10.21227/n69k-
500 r458). The source code is available at [https://github.com/mmann1123/South_](https://github.com/mmann1123/South_Africa_Crop_Comp)
501 [Africa_Crop_Comp](https://github.com/mmann1123/South_Africa_Crop_Comp) (Mann, 2025a). The study builds on the ground-reference
502 crop labels of the Western Cape Department of Agriculture and Radiant Earth
503 Foundation, “Crop Type Classification Dataset for Western Cape, South Africa”,
504 v1.0, Radiant MLHub (DOI: 10.34911/rdnt.j0co8q), used under the CC-BY-4.0
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